

Gemstone Quantitative Suite

A Multi-Asset Regime Detection & Execution Framework

Technical Summary: AGATE · BERYL · CITRINE

Perry Jenkins

March 2026

Abstract

We present three autonomous trading systems unified by a common mathematical core: N -state Gaussian Hidden Markov Models for latent regime detection in financial time series. The suite exploits the analogy between market regimes and metastable states in statistical mechanics — markets occupy quasi-stationary configurations (BULL, BEAR, CHOP) with stochastic transitions governed by a row-stochastic matrix A . Each system layers domain-specific innovations atop this shared foundation: AGATE applies per-ticker ensemble HMMs to 14 cryptocurrency futures on 3–4 hour bars; BERYL implements multi-position equity rotation across 82 NDX100 tickers with SEC Form 4 insider data as a Bayesian prior; CITRINE constructs confidence-weighted portfolios with risk-parity scaling and adaptive cash bands. All three are validated via walk-forward analysis with chained equity and deployed 24/7 on cloud infrastructure with real-money execution.

Contents

1. Executive Overview
2. Mathematical Foundation: Latent Regime Detection via Gaussian HMM
3. Ensemble Consensus: Robustness via Model Averaging
4. Signal Generation: Multi-Gate Confirmation Filter
5. System-Specific Innovations
6. Walk-Forward Validation Framework
7. Shared Infrastructure & Deployment
8. Statistical Properties & Open Questions

1. Executive Overview

The Gemstone Suite comprises three autonomous trading systems operating across distinct asset classes, unified by a common mathematical core: N -state Gaussian Hidden Markov Models (HMMs) for latent regime detection, ensemble consensus mechanisms for robustness, and multi-gate confirmation filters for signal purity. Each system is deployed 24/7 on Oracle Cloud Infrastructure with real-money execution.

System	Asset Class	Universe	Timeframe	Key Innovation
AGATE	Crypto futures (Coinbase)	14 tickers (BTC, ETH, SOL, ...)	3–4h bars	Per-ticker ensemble HMM with walk-forward grid opt.
BERYL	US equities (NDX100)	82 tickers	Daily bars	Multi-position rotation with SEC Form 4 insider prior
CITRINE	US equities (NDX100)	82 tickers	Daily bars	Confidence-weighted alloc. with risk-parity & adaptive cash

Table 1: System overview. All three share a common trading-core library.

The shared library (trading-core/) provides HMM fitting, 8-indicator confirmation, event-driven backtesting, and walk-forward validation. Each system adds domain-specific allocation, risk management, and execution logic.

2. Mathematical Foundation: Latent Regime Detection

2.1 The Generative Model

We model the observed market microstructure as emissions from a latent Markov chain. Let $S_t \in \{1, \dots, K\}$ denote the hidden regime state at time t , and $\mathbf{x}_t \in \mathbf{R}^d$ the observation vector. The joint probability factorizes as:

$$P(\mathbf{x}_{1:T}, S_{1:T}) = \pi_{S_1} \cdot \prod_{t=2..T} A_{S_{t-1}, S_t} \cdot \prod_{t=1..T} \phi(\mathbf{x}_t; \mu_{S_t}, \Sigma_{S_t})$$

where:

- $\pi \in \Delta^{K-1}$: initial state distribution on the $(K-1)$ -simplex
- $\mathbf{A} \in \mathbf{R}^{K \times K}$: row-stochastic transition matrix encoding regime persistence and transition rates
- $\phi(\cdot; \mu_k, \Sigma_k)$: Gaussian emission density for state k

The transition matrix $\mathbf{A}$ encodes regime "stickiness" — diagonal elements A_{kk} represent regime persistence (analogous to metastable states in statistical mechanics with mean lifetime $\tau_k = 1/(1 - A_{kk})$), while off-diagonal elements capture phase transition rates. Empirically, $A_{kk} > 0.85$, yielding mean regime persistence of ~7 bars.

2.2 Feature Space Construction

The observation vector $\mathbf{x}_t$ is constructed from OHLCV data through continuous transformations. A critical architectural constraint: **all features must be continuous-valued**. Binary features (e.g., {0,1} divergence flags) violate the Gaussian emission assumption and produce degenerate covariance matrices when $K \geq 5$ — discovered empirically when eigenvalue collapse caused numerical instability.

Feature	Definition	Interpretation
r_t	$\ln(C_t / C_{t-1})$	Log-return (additive under composition)
ρ_t	$(H_t - L_t) / C_t$	Normalized price range (intraday volatility proxy)
v_t	$V_t / V_{t-1} - 1$	Relative volume change (conviction signal)
σ -ratio	$\sigma(r, \tau_{\text{short}}) / \sigma(r, \tau_{\text{long}})$	Realized vol ratio (vol-of-vol regime)
α_1	$\text{Corr}(r_t, r_{t-1})$ over window	Lag-1 autocorrelation (mean-reversion signal)
κ	Fisher kurtosis of r over window	Excess kurtosis (fat-tail / jump-risk)

Table 2: Feature space. Base set uses top 3; extended_v2 uses all 6. All features winsorized to [1st, 99th] percentiles and standardized.

2.3 Parameter Estimation: Baum-Welch Algorithm

Parameters $\theta = \{\pi, \mathbf{A}, \{\mu_k, \Sigma_k\}_{k=1..K}\}$ are estimated via the Baum-Welch algorithm (Expectation-Maximization for HMMs):

E-step: Compute forward-backward variables $\alpha_t(k), \beta_t(k) \rightarrow$ posterior $\gamma_t(k) = P(S_t = k | \mathbf{x}_{1:T})$

M-step: Update as weighted MLEs:

$$\begin{aligned} \mu_k &\leftarrow \sum_t \gamma_t(k) \mathbf{x}_t / \sum_t \gamma_t(k) \\ \Sigma_k &\leftarrow \sum_t \gamma_t(k) (\mathbf{x}_t - \mu_k)(\mathbf{x}_t - \mu_k)^T / \sum_t \gamma_t(k) \\ A_{jk} &\leftarrow \sum_t \xi_t(j, k) / \sum_t \gamma_t(j) \end{aligned}$$

Robustness measures:

- **Multiple restarts:** $n_{\text{init}} = 3$ random initializations; retain maximum log-likelihood solution. Mitigates EM sensitivity to initial conditions (local optima in the likelihood landscape).
- **Covariance regularization:** If $\lambda_{\min}(\Sigma_k) < 10^{-6}$, apply $\Sigma_k \leftarrow \Sigma_k + 10^{-4} \cdot \mathbf{I}$. This Tikhonov regularization is analogous to adding a thermal noise floor — it prevents measure-zero concentrations in the emission density.
- **Convergence guard:** Reject models where any state has < 5 assigned observations. Prevents "ghost states" with ill-defined emission distributions.

2.4 Regime Labeling: Data-Driven State Classification

Post-fitting, states are labeled without hardcoded assumptions. This is analogous to labeling energy eigenstates by their eigenvalues — the labels emerge from the data:

- Compute empirical mean return per state via Viterbi path: $\hat{\mu}_k = (1/n_k) \sum r_t$ for all t assigned to state k
- Sort states by $\hat{\mu}_k$ in descending order
- Assign top $N_{\text{bull}} \rightarrow \text{BULL}$, bottom $N_{\text{bear}} \rightarrow \text{BEAR}$, middle $\rightarrow \text{CHOP}$ (sub-ranked by $\text{Tr}(\Sigma_k)$ as volatility proxy)

2.5 Inference

State prediction uses Viterbi decoding (MAP sequence) with posterior confidence $\hat{c}_t = \max_k P(S_t = k \mid \mathbf{x}_{1:T})$. This posterior probability becomes the primary input to downstream allocation, acting as a continuous-valued "conviction" measure for the detected regime.

3. Ensemble Consensus: Robustness via Model Averaging

3.1 The Overfitting Problem

Single- K HMMs exhibit extreme sensitivity to the state-space dimension. Empirical finding across the suite:

$$K=5: \text{ Sharpe} = -3.567 \qquad K=6: \text{ Sharpe} = +5.063 \qquad K=7: \text{ Sharpe} = -0.309$$

This parametric fragility — reminiscent of the sensitivity of partition function calculations to the number of included microstates — motivates an ensemble approach.

3.2 Architecture

Three HMMs with $K \in \{K^*-1, K^*, K^*+1\}$ are fitted independently. At inference, for each bar t :

- Each model $m \in \{1,2,3\}$ predicts regime category $c_t^{(m)} \in \{\text{BULL}, \text{BEAR}, \text{CHOP}\}$
- Majority vote: if ≥ 2 models agree $\rightarrow$ winner = consensus regime
- Consensus confidence = mean posterior of agreeing models
- If all three disagree $\rightarrow$ conservative fallback to CHOP

This is analogous to a **quorum-based Byzantine fault tolerance protocol** — requiring 2-of-3 agreement filters out single-model overfitting artifacts while preserving genuine regime signals. The ensemble reduces the standard deviation of walk-forward Sharpe by $\sim 40\%$ compared to any single- K model, at the cost of $\sim 15\%$ reduction in peak Sharpe — the classic bias-variance tradeoff.

3.3 Graceful Degradation

If fewer than 2 models converge (per the min-observations guard), the system falls back to a single 4-state HMM (minimum viable model). This ensures the system always produces predictions, even with pathological data.

4. Signal Generation: Multi-Gate Confirmation Filter

4.1 The 8-Indicator Gate

Beyond HMM regime detection, entry signals must pass through 8 independent technical indicator confirmations operating as a conjunction filter. Each indicator i returns $b_i \in \{0,1\}$:

Gate	Indicator	LONG Condition	SHORT Condition
1	RSI(14)	$30 < \text{RSI} < 70$	$30 < \text{RSI} < 70$
2	ROC(10)	$\text{ROC} > 0$	$\text{ROC} < 0$
3	$\sigma(20)$	$\sigma < 2 \cdot \text{median}(\sigma)$	$\sigma < 2 \cdot \text{median}(\sigma)$
4	$V / \text{SMA}(V,20)$	$\text{ratio} > 1.1$	$\text{ratio} > 1.1$
5	ADX(14)	$\text{ADX} > 20$	$\text{ADX} > 25$ (asymmetric)
6	C vs SMA(50)	$C > \text{SMA}$	$C < \text{SMA}$
7	MACD(12,26,9)	$\text{MACD} > \text{signal}$	$\text{MACD} < \text{signal}$
8	%K Stoch(14,3)	$\%K < 80$	$\%K > 20$

Table 3: Confirmation gate specification. Entry requires $\sum b_i \geq \tau_{\text{confirm}}$ (per-ticker, typically 5–7).

The asymmetric ADX gate (20 for LONG, 25 for SHORT) reflects the empirical observation that shorting requires stronger trend confirmation — a regularization on the more dangerous direction.

4.2 Information-Theoretic Interpretation

Each indicator gate can be viewed as an independent binary channel with its own noise characteristics. The conjunction requirement (≥ 7 of 8) implements a low-false-positive decoder. The probability of 7+ independent false positives simultaneously is:

$$P(\sum b_i \geq 7) = C(8, 7) \cdot p^7(1-p) + p^8 \approx 8p^7$$

For per-indicator false positive rate $p = 0.3$, this gives ≈ 0.002 — a **500× noise reduction**. This is the core mechanism by which the system achieves high precision despite imperfect individual indicators.

5. System-Specific Innovations

5.1 AGATE: Crypto Futures Regime Trading

Asset class: 14 cryptocurrency futures on Coinbase Advanced (BTC, ETH, SOL, XRP, LTC, SUI, DOGE, ADA, BCH, LINK, HBAR, XLM, AVAX, ENA)

Timeframe: 3–4 hour bars (intraday regime detection)

Capital: \$2,500 max notional per ticker, 0.25× leverage

Key Innovations

- **Per-ticker hyperparameter optimization:** Walk-forward grid search over 144 combinations per ticker ($K \times \text{feature_set} \times \text{confirmations} \times \text{cov_type} \times \text{timeframe}$). 6-month train / 3-month test rolling windows with chained equity curves.
- **Stability-penalized scoring:** $\text{rank_score} = \text{Sharpe}_{\text{WF}} \times \text{consistency}$, where consistency = (positive windows)/(total windows). Penalizes parameter sets that are spectacular in one window but catastrophic in others — analogous to requiring low variance in a Monte Carlo estimator.
- **Dominant patterns:** Diagonal covariance universally preferred; 4-hour timeframe dominates; extended_v2 features outperform base by +2.75 Sharpe on average.

Ticker	K	Features	Confirms	WF Sharpe	Return	Consistency
ETH	5	extended	5	5.06	+22.0%	1/2
LINK	4	base	5	4.13	+22.2%	4/5
BCH	5	ext_v2	5	3.52	+20.0%	5/5 ✓
SOL	4	ext_v2	8	2.89	+2.4%	2/2
XLM	6	extended	6	2.15	+42.8%	4/5

Table 4: AGATE walk-forward performance. BCH is the stability champion (5/5 positive windows).

5.2 BERYL: Equity Rotation with Alternative Data

Asset class: 82 NDX100 equities

Timeframe: Daily bars

Execution: Multi-position rotation (up to 3 simultaneous longs, ~\$833 each)

Position Scoring Formula

$\text{score} = \text{confidence} \times (\text{confirmations} / \text{min_confirms}) \times \text{alt_boost} \times \text{persistence_bonus}$

where:

- **alt_boost:** SEC Form 4 insider trading data as multiplicative Bayesian prior [0.7, 1.0, 1.15]. 10b5-1 pre-scheduled sales are filtered out; only discretionary insider *buying* within 90 days generates a positive signal.
- **persistence_bonus:** $1.0 + \min(\max(\text{days} - 3, 0) \times 0.10, 0.50)$. Rewards established regimes (>13 days → 1.5×); penalizes fresh flips. Acts as momentum confirmation.

Kill-Switch Logic

- Total P&L loss > 2% of equity → halt all trading

- Zero wins in last 10 trades → halt
- Rolling 5-trade Sharpe < 0.3 → halt

Performance: Walk-forward Sharpe 0.7–0.9 on top tickers. First live trade: +2.77% on ARM.

5.3 CITRINE: Confidence-Weighted Portfolio Allocation

Asset class: 82 NDX100 equities (same universe as BERYL, different strategy)

Timeframe: Daily bars

Execution: Up to 15 simultaneous positions with risk-parity weighting

Citrine Score (Portfolio Weight Input)

$$w_i = f(c_i^{\wedge}) \cdot (\sigma_{\text{median}} / \sigma_i) \cdot (\sum_j b_{ij} / 8) \cdot g(\text{persistence}_i)$$

where:

- $f(c^{\wedge}) = \text{clip}((c^{\wedge} - 0.65) / 0.25, 0, 1)$ — confidence mapping with dead zone
- $\sigma_{\text{median}} / \sigma_i$ — inverse-volatility (risk parity) scaling, clipped to [0.5, 2.0]. More volatile assets receive proportionally lower weight.
- $\sum b_{ij} / 8$ — indicator quality fraction (continuous [0, 1])
- $g(p) = 1 + \min(\max(p-3, 0) \times 0.1, 0.5)$ — persistence bonus

Hysteresis Bands (Schmitt Trigger Analogy)

Entry and exit use asymmetric thresholds to prevent chattering near the decision boundary:

- Entry: $c^{\wedge} \geq 0.90$ AND persistence ≥ 3 days
- Hold: $c^{\wedge} \geq 0.50$
- Dead zone [0.50, 0.90]: hold existing positions, don't enter new ones

This is directly analogous to a **Schmitt trigger** in electronics — the hysteresis band prevents rapid oscillation when the input hovers near a threshold. In the language of dynamical systems, it introduces a potential barrier between the "invested" and "uninvested" states.

Adaptive Cash Positioning (Market-Breadth Risk Gauge)

The fraction of capital held in cash is dynamically determined by the number of tickers detected in BULL regime across the entire universe:

N_bull	Cash %	Invested %	Interpretation
≥ 15	10%	90%	Broad strength → risk-on
8–14	15%	85%	Moderate breadth
3–7	25%	75%	Narrow leadership
0–2	50%	50%	Defensive → risk-off

Table 5: Adaptive cash bands. Acts as an emergent market-breadth indicator.

Walk-Forward Performance

Window	Period	Return	QQQ B&H	Alpha	Sharpe	Max DD
1	Nov 24 – Feb 25	+2.84%	+1.77%	+1.07%	1.67	-1.59%
2	May 25 – Aug 25	-1.46%	+17.74%	-19.20%	-0.95	-3.48%
3	Nov 25 – Feb 26	+5.37%	-2.18%	+7.55%	3.30	-1.12%

Table 6: CITRINE walk-forward results. Generates alpha in multi-regime environments; underperforms in monotonic bull markets (expected for regime-switching strategies).

6. Walk-Forward Validation Framework

All three systems use identical out-of-sample validation with **chained equity** — the terminal equity of window w becomes the initial equity of window $w+1$, preventing the "fresh start" bias that masks compounding effects of losses.

Protocol:

- For each rolling window w : train HMM on $[t_w - \tau_{\text{train}}, t_w)$ — strict temporal separation, no look-ahead
- Predict regimes on $[t_w, t_w + \tau_{\text{test}})$ — true out-of-sample
- Execute strategy, record equity curve
- Chain equity: $E_{w+1,0} = E_{w,T}$

Metrics per window:

$$\text{Sharpe} = (\mu_r / \sigma_r) \times \sqrt{\text{annualization factor}}$$

$$\text{Max DD} = \min_t (E_t - \max_{s \leq t} E_s) / \max_{s \leq t} E_s$$

$$\text{Alpha} = R_{\text{portfolio}} - R_{\text{benchmark}}$$

The annualization factor varies by system: $\sqrt{24 \times 365}$ for AGATE (hourly crypto), $\sqrt{252}$ for BERYL/CITRINE (daily equities).

7. Shared Infrastructure & Deployment

7.1 Common Library

Module	Purpose
hmm_model.py	N-state Gaussian HMM with auto-labeling, covariance regularization
ensemble.py	3-model majority voting ensemble (2-of-3 consensus)
indicators.py	8 technical indicators + extended HMM features
strategy.py	Signal generation: LONG/SHORT entry/exit logic
backtester.py	Event-driven bar-by-bar execution simulator
walk_forward.py	Rolling train/test validation harness
alternative_data.py	SEC Form 4 insider filing parser

Table 7: Shared trading-core library modules.

7.2 Deployment

- Oracle Cloud Infrastructure compute instance (Ubuntu, 24/7)
- Each system runs as independent process with SQLite persistence
- Pushover alerts for trades, errors, kill-switch activations
- Data sources: Polygon.io (equities), Coinbase Advanced (crypto)
- Per-ticker optimized configs stored as JSON; updated via walk-forward grid search

8. Statistical Properties & Open Questions

8.1 Empirical Observations

Regime persistence: Diagonal elements of $\mathbf{A}$ typically satisfy $A_{kk} > 0.85$, yielding mean regime lifetimes of ~ 7 bars ($1/[1 - A_{kk}]$). This is consistent with the well-documented clustering of volatility in financial time series (Mandelbrot, 1963; Engle, 1982).

Feature set impact: `extended_v2` (including realized kurtosis κ) consistently outperforms base features by $+0.5$ – 2.5 Sharpe across all three systems. The kurtosis feature captures jump-risk regimes that pure return-based features miss — analogous to including higher-order cumulants beyond the mean-field (Gaussian) approximation.

Ensemble value-add: The 2-of-3 consensus reduces Sharpe standard deviation by $\sim 40\%$ vs. any single- K model, at the cost of $\sim 15\%$ reduction in peak Sharpe. This is the bias-variance tradeoff in its canonical form.

8.2 Open Questions

- **Semi-Markov extensions:** Can regime transition rates be modeled as time-varying to capture accelerating transitions during market stress? The current homogeneous Markov assumption implies exponentially distributed sojourn times, which may underestimate the probability of rapid regime cascades.
- **Cross-asset coupling:** Would a joint HMM across correlated tickers (shared latent state) improve signal quality? This is equivalent to moving from mean-field (independent) to coupled-oscillator dynamics.
- **Non-Gaussian emissions:** The Gaussian assumption may be violated during fat-tailed events. Student- t emissions with learnable degrees of freedom ν would provide heavier tails and potentially better tail-risk detection.
- **Continuous-time limit:** As bar resolution increases ($4h \rightarrow 1h \rightarrow \text{tick}$), does the discrete HMM converge to a continuous-time regime-switching diffusion (Hamilton, 1989)? This would connect the framework to the established theory of jump-diffusion processes.
- **Information geometry:** The space of HMM parameters forms a statistical manifold. Do walk-forward-optimal parameters cluster in specific regions of this manifold? If so, the Fisher information metric could guide more efficient optimization.